

Broadcom AutoSys

Enterprise job scheduling and workload automation in plain English

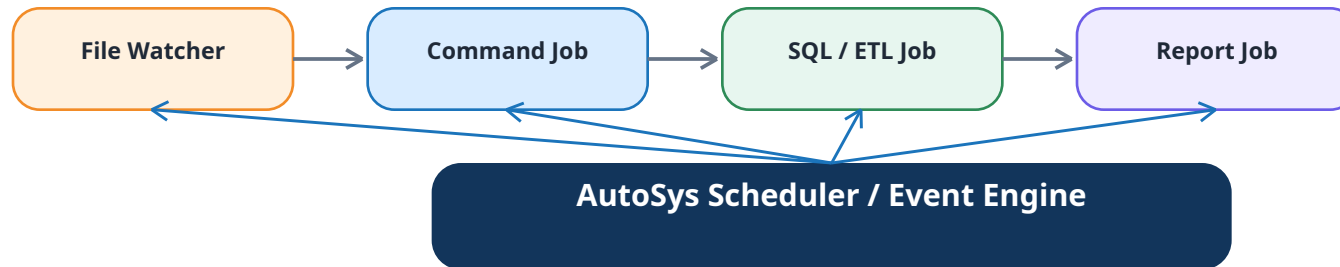
1 What AutoSys is really doing

Broadcom AutoSys Workload Automation is an enterprise scheduler and workload orchestration platform. In data engineering and operations, it coordinates scheduled jobs, dependencies, calendars, file-watchers, scripts, database work, ETL steps, and production batch flows across many servers.

- Schedule jobs by date, time, calendar, and business rules
- Start jobs only when dependency conditions are satisfied
- Run work on target machines through AutoSys agents
- Track status, alerts, reruns, history, and operational risk

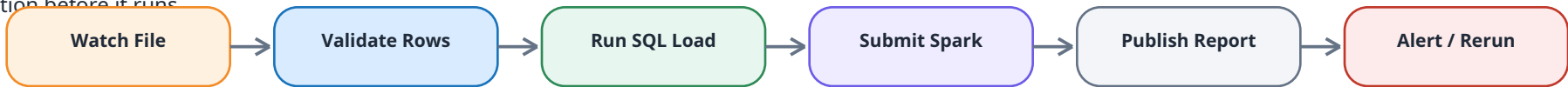
AutoSys = job traffic control

It coordinates the flow. The real processing still happens in scripts, SQL engines, Spark, ETL tools, applications, and file-transfer systems.

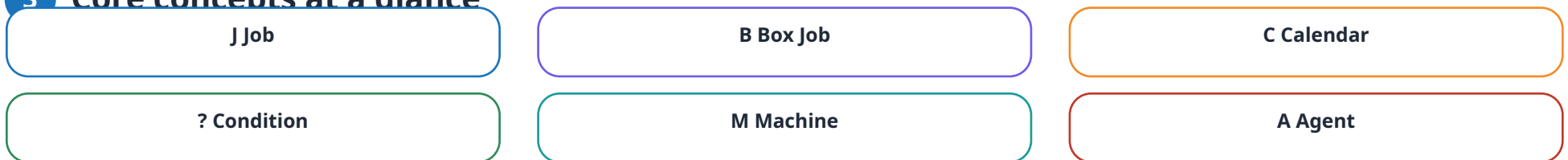


2 The dependency flow idea

AutoSys workflows are built from jobs connected by starting conditions. A job can wait for another job, a box, a file, a calendar, a time window, or a machine/resource condition before it runs.



3 Core concepts at a glance



AutoSys Architecture

How the scheduler, event server, clients, and agents work together

4 Six key components

1 Web UI / WCC / Client Tools

Define jobs, inspect flows, hold/start jobs, view output, and monitor production.

2 Application Server

Communication layer between clients and the event server.

3 Scheduler

The scheduling engine. It evaluates events, job definitions, calendars, and dependencies.

4 Event Server / Database

Stores definitions, events, status, calendars, machines, history, and state.

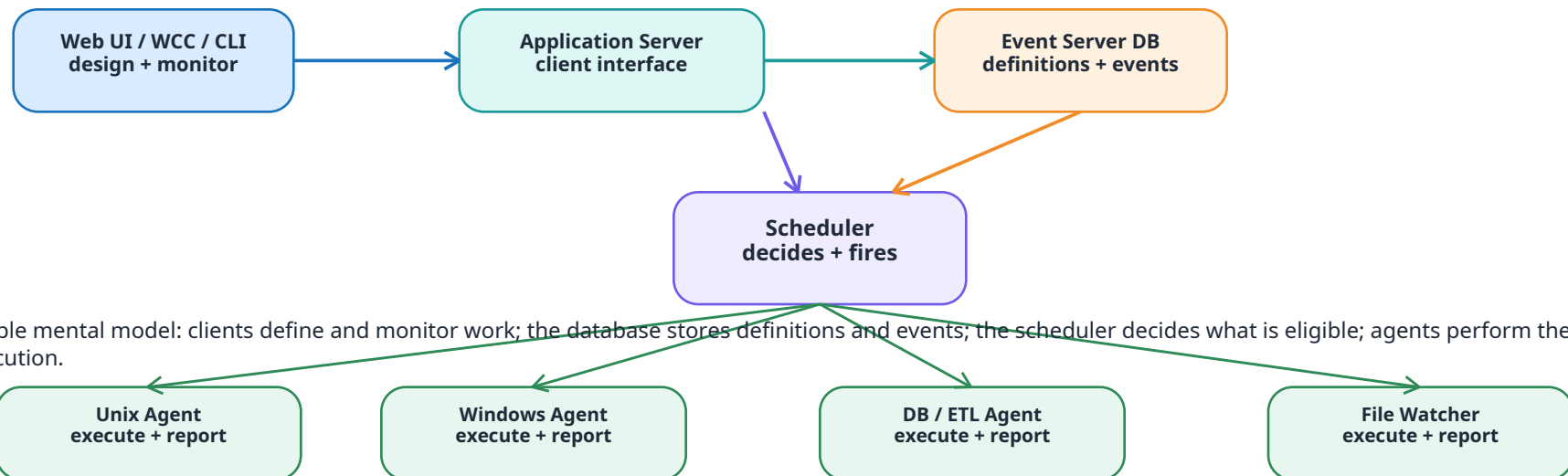
5 Agents

Run commands or integrations on target machines and report status back.

6 Monitoring and Alerts

Failures, late jobs, reruns, notifications, dashboards, and support response.

5 Architecture data flow

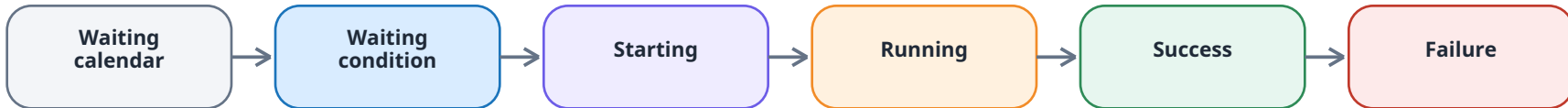


Simple mental model: clients define and monitor work; the database stores definitions and events; the scheduler decides what is eligible; agents perform the actual execution.

Running Workloads

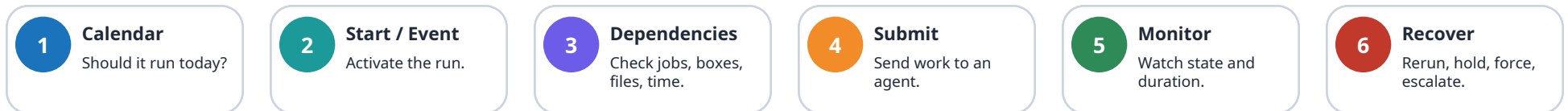
Schedules, states, job types, and a jobs-as-code pattern

6 Job lifecycle in operations language



Scheduler decides eligibility. Agent executes. Event server remembers status, history, events, output references, reruns, and alerts.

7 How scheduling usually works



8 A simple AutoSys JIL-style pattern

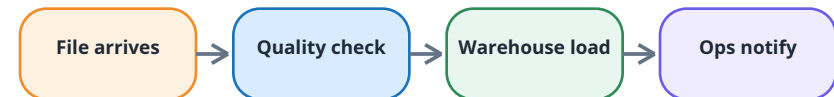
Jobs-as-code style example

```
insert_job: daily_customer_load  job_type: BOX
date_conditions: 1
run_calendar: business_days
start_times: "06:00"
```

```
insert_job: extract_file  job_type: FW
box_name: daily_customer_load
watch_file: /drop/customer_*.csv
```

```
insert_job: validate_rows  job_type: CMD
condition: success(extract_file)
command: python validate_rows.py
```

Do not get stuck on exact syntax. The key idea is that AutoSys knows the business calendar, dependency chain, target machine, success rule, and recovery path.



9 Common job types you will hear about



How to Think About AutoSys

Use cases, boundaries, habits, and interview language

10 AutoSys is / AutoSys is not

AutoSys is

- An enterprise job scheduler
- A workload automation platform
- A dependency manager across systems
- A monitor for job state, alerts, and history

AutoSys is not

- The main data processing engine
- A replacement for SQL, Spark, Python, or ETL logic
- A low-latency streaming platform
- A magic fix for fragile pipeline design

11 Best use cases



Batch ETL / ELT

Nightly loads, warehouse refreshes, marts, extracts.



Enterprise Ops

Unix, Windows, databases, and apps.



File Workflows

File arrival, watchers, validation, transfers.



SLA Workflows

Critical paths, late jobs, recovery.



Hybrid Estates

Older systems plus newer platforms.

12 Production habits that matter

- Name jobs, boxes, calendars, machines, and conditions clearly for support teams.
- Separate orchestration from processing: AutoSys schedules; scripts, SQL, Spark, and apps process.
- Use calendars, dependencies, and file watchers instead of fragile sleep commands.
- Capture logs, rerun instructions, ownership, alert routing, and normal runtime expectations.
- Treat critical workflows as products: version definitions, test changes, and maintain runbooks.

13 Interview-ready explanation

Broadcom AutoSys is an enterprise workload automation platform. I think of it as job traffic control for production workflows: it schedules jobs, evaluates calendars and dependencies, sends work to agents on target machines, monitors success or failure, and helps operations respond to alerts and SLA risk. The heavy processing still happens in SQL, Spark, scripts, ETL tools, applications, or file-transfer systems - AutoSys coordinates the flow.